The McConnell Bond-Anisotropy Calculator

Generated/Reviewed Documentation (Codex/Opus/Human) for v0.0.1

1 What the calculator computes

The McConnell calculator describes the through-space magnetic effect of a covalent bond whose electronic response is different along the bond axis than across it. In a magnetic field, that anisotropic response creates a secondary field at a nearby nucleus. The calculator represents the geometric part of that effect as a rank-2 tensor: a 3×3 array whose ab entry says how a field component applied along direction b contributes to the induced field component along direction a.

In the code, each accepted bond near each target atom contributes a tensor with units Å^{-3}. The factor that would turn this geometric tensor into a shielding contribution, such as a bond susceptibility anisotropy in the usual McConnell formula, is not applied here. The output can therefore correlate with a DFT shielding contribution after calibration, but before calibration it is a geometric descriptor rather than a chemical shift or a shielding in ppm.

2 Where shielding enters

For a nucleus in an applied magnetic field $\boldsymbol{B}_0$, the local field is usually written

$$\boldsymbol{B}_{\text{nuc}} = (\boldsymbol{I} - \boldsymbol{\sigma})\boldsymbol{B}_0. \tag{1}$$

Here $\boldsymbol{I}$ is the 3×3 identity matrix and $\boldsymbol{\sigma}$ is the nuclear shielding tensor. The tensor is dimensionless in physical NMR: it is the linear-response coefficient connecting the applied field to the field screened by the electrons. The matrix form is needed because a molecule is not a sphere. Rotating the molecule can change both the size and the direction of the secondary field at the nucleus.

Equation (1) is a quantum-mechanical statement about electrons: the applied field perturbs the electronic state, the perturbed electrons produce their own magnetic field at the nucleus, and the shielding tensor records the linear part of that response. The McConnell calculator keeps a classical far-field approximation to one part of this response. It treats a bond as a small anisotropic magnetic object located at its midpoint. The bond direction sets the anisotropy axis, and the vector from the bond midpoint to the target atom sets the observation direction. This is a local geometric model of the angular response; it does not solve the electronic structure problem.

The NMR observable most often reported in solution is the isotropic chemical shift. It is related to the isotropic shielding, the average of the three diagonal tensor entries, after reference subtraction and sign conventions. That scalar does not exhaust the calculation. The orientation-dependent part of $\boldsymbol{\sigma}$, carried by the symmetric traceless tensor components, is retained because it records how the local geometry affects the angular pattern of the shielding response.

3 The tensor split used by the code

A real 3×3 tensor $\boldsymbol{X}$ is stored through a closed-form split. The C++ type `SphericalTensor` owns this split. Its `Decompose` routine is implemented in `Types.cpp`. The scalar component is

$$T_0 = \frac{1}{3} \operatorname{tr} \boldsymbol{X}. \tag{2}$$

This is the isotropic part: it multiplies the identity matrix and survives orientational averaging.

The antisymmetric part is stored as a three-component pseudovector,

$$\boldsymbol{T}_1 = \frac{1}{2}\begin{pmatrix} X_{yz} - X_{zy} \\ X_{zx} - X_{xz} \\ X_{xy} - X_{yx} \end{pmatrix}. \tag{3}$$

It is a compact way to store the matrix part that changes sign under transpose. For standard isotropic solution NMR this part is usually not observed directly, but the full calculator can produce it because the McConnell tensor is generally not symmetric.

The remaining symmetric traceless matrix is

$$\boldsymbol{S} = \frac{\boldsymbol{X} + \boldsymbol{X}^{\mathsf{T}}}{2} - T_0\boldsymbol{I}, \qquad \operatorname{tr}\boldsymbol{S} = 0. \tag{4}$$

It has five independent entries. The code packs them into the real spherical T_2 basis

$$\boldsymbol{T}_2 = \begin{pmatrix} \sqrt{2}\,S_{xy} \\ \sqrt{2}\,S_{yz} \\ \sqrt{3/2}\,S_{zz} \\ \sqrt{2}\,S_{xz} \\ (S_{xx} - S_{yy})/\sqrt{2} \end{pmatrix}. \tag{5}$$

The factors in Eq. (5) are the code's normalization. They preserve the Frobenius norm: the squared length of this five-vector equals $\sum_{ab} S_{ab}^2$. The nine stored doubles are ordered as

$$[T_0, T_{1x}, T_{1y}, T_{1z}, T_{2,-2}, T_{2,-1}, T_{2,0}, T_{2,+1}, T_{2,+2}].$$

For McConnell, the overall bond total is decomposed into the three parts. The category tensor outputs are projected to the T_2 subspace before they are written.

4 The method implemented

For a target atom at position $\boldsymbol{r}_i$ and a covalent bond j with endpoint positions $\boldsymbol{q}_A$ and $\boldsymbol{q}_B$, the geometry module provides

$$\boldsymbol{m}_j = \frac{\boldsymbol{q}_A + \boldsymbol{q}_B}{2}, \qquad \hat{\boldsymbol{b}}_j = \frac{\boldsymbol{q}_B - \boldsymbol{q}_A}{|\boldsymbol{q}_B - \boldsymbol{q}_A|}. \tag{6}$$

The midpoint $\boldsymbol{m}_j$ has units of Å, while $\hat{\boldsymbol{b}}_j$ is a unitless bond-axis vector. Reversing the endpoint order does not change the final tensor, because both appearances of $\hat{\boldsymbol{b}}_j$ change sign where needed.

The displacement from the bond midpoint to the target atom is

$$\boldsymbol{x}_{ij} = \boldsymbol{r}_i - \boldsymbol{m}_j, \qquad r_{ij} = |\boldsymbol{x}_{ij}|, \qquad \hat{\boldsymbol{d}}_{ij} = \frac{\boldsymbol{x}_{ij}}{r_{ij}}, \qquad c_{ij} = \hat{\boldsymbol{d}}_{ij} \cdot \hat{\boldsymbol{b}}_j. \tag{7}$$

The distance r_{ij} is in Å; c_{ij} is the cosine of the angle between the bond axis and the midpoint-to-atom direction.

The full tensor contribution stored by the implementation is

$$\mathcal{M}_{ab}^{(ij)} = \frac{9c_{ij}\,\hat{d}_{ij,a}\hat{b}_{j,b} - 3\hat{b}_{j,a}\hat{b}_{j,b} - \left(3\hat{d}_{ij,a}\hat{d}_{ij,b} - \delta_{ab}\right)}{r_{ij}^3}. \tag{8}$$

The Kronecker delta δ_{ab} is one when $a = b$ and zero otherwise. The Cartesian indices a and b run over x, y, z. The numerator is dimensionless, so $\mathcal{M}^{(ij)}$ has units Å^{-3}. The first dyadic product, $\hat{\boldsymbol{d}} \otimes \hat{\boldsymbol{b}}$, is generally asymmetric and can therefore produce T_1. The second term is a projection onto the bond axis. The third term is the negative of a symmetric traceless dipolar tensor.

For each accepted bond, the code also stores the dipolar tensor

$$K_{ab}^{(ij)} = \frac{3\hat{d}_{ij,a}\hat{d}_{ij,b} - \delta_{ab}}{r_{ij}^3} \tag{9}$$

and the scalar

$$f_{ij} = \frac{3c_{ij}^2 - 1}{r_{ij}^3}. \tag{10}$$

Both $K^{(ij)}$ and f_{ij} have units Å^{-3}. The angular factor $3c^2 - 1$ is the same two-lobed dipolar pattern carried by the T_2 tensor: it is positive near the bond axis, negative near directions perpendicular to the bond, and zero at $c = 1/\sqrt{3}$, the magic angle of about $54.7°$. The scalar is the double contraction $\hat{\boldsymbol{b}}_j^{\top}\boldsymbol{K}^{(ij)}\hat{\boldsymbol{b}}_j$. For the full tensor in Eq. (8), $\operatorname{tr}\mathcal{M}^{(ij)} = 3f_{ij}$, so f_{ij} is the T_0 value of a single-bond full tensor, not the trace of the dipolar tensor K, whose trace is zero.

A scalar textbook McConnell term is often written with a prefactor such as $(\Delta\chi/3)f_{ij}$, where $\Delta\chi$ is the bond susceptibility anisotropy. This implementation stops before that multiplication. It stores $\mathcal{M}^{(ij)}$, $K^{(ij)}$, and f_{ij} as unscaled geometry, and it does not choose separate susceptibility constants for C=O, C–N, aromatic, or other bonds.

For each atom, the calculator queries bond midpoints within 10.0 Å. This is the McConnell bond-anisotropy cutoff in `data/calculator_params.toml`. A candidate contribution is rejected when $r_{ij} < 0.1$ Å, when the target atom is one of the bond endpoints, or when $r_{ij} \leq 0.5\,\ell_j$, where ℓ_j is the bond length. The 0.1 Å guard prevents numerical divergence of the r^{-3} expression. The $0.5\,\ell_j$ rule excludes points inside the near-field region of the finite bond source, where a point-source dipole description is not the model being evaluated.

Accepted tensors are accumulated as

$$\mathcal{M}_i = \sum_{j\in\mathcal{A}_i} \mathcal{M}^{(ij)}, \tag{11}$$

where $\mathcal{A}_i$ is the set of accepted nearby bonds for atom i. $\mathcal{M}_i$ is decomposed by Eq. (2)–(5) and written as the nine-column full McConnell output. The source code field is named `mc_shielding_contribution`; despite that name, the stored values are the unscaled Å^{-3} geometric tensor.

The calculator also keeps category summaries. The scalar f_{ij} is summed separately for peptide C=O bonds, peptide C–N bonds, side-chain C=O bonds, and aromatic bonds. Full tensors are summed into three broader groups: backbone bonds, side-chain bonds, and aromatic bonds. Before writing those group tensors as category features, the code symmetrizes each group tensor and subtracts its trace divided by three, so those outputs contain the five T_2 components and no T_0 or T_1 part. The nearest peptide C=O and nearest peptide C–N bonds are selected by midpoint distance; their stored T_2 descriptors come from the dipolar tensor K in Eq. (9). Disulfide and unclassified bonds still enter the overall total after passing the filters, but they do not enter the written category T_2 groups or scalar sums.

The main approximations are therefore visible in the code path. A bond is collapsed to its midpoint and unit axis. Accepted bonds have unit geometric weight in the base calculator; no bond-order, element, or susceptibility coefficient is multiplied into Eq. (8). The r^{-3} form is used outside a fixed near-field exclusion, and the category labels are used for output grouping rather than for changing the formula.

The same tensor expression can also be sampled at an arbitrary point in space. That path iterates over the bond list, accepts bonds within 10.0 Å, applies the 0.1 Å singularity guard, and omits points with $r_{ij} < 0.5\,\ell_j$. The per-atom path requires $r_{ij} > 0.5\,\ell_j$, so it rejects equality at that near-field boundary as well. The free-point path does not run the endpoint self-source test, because a sampling point has no atom index. When the nearest peptide C=O direction is stored on the atom, the code re-normalizes it unless its norm is below 10^{-10}, in which case the direction is written as zero.

5 Inputs and output

This calculator consumes molecular geometry: atom positions in Å, typed covalent bonds, bond categories, bond midpoints, bond lengths, and bond directions. Its declared runtime dependencies are the geometry result and the spatial index over bond midpoints. It does not consume MOPAC charges, AIMNet2 charges or embeddings, ff14SB charges, APBS electrostatic fields, or DSSP secondary-structure data. Those inputs feed other charge, electrostatic, and secondary-structure calculations in the pipeline. MOPAC bond orders are used by the separate MOPAC-weighted McConnell variant, not by this calculator.

The feature files reflect that geometry-derived role. `mc_shielding.npy` stores an $N \times 9$ table: the nine $T_0 + T_1 + T_2$ components of $\mathcal{M}_i$ for each atom. `mc_category_T2.npy` stores an $N \times 25$ table with five five-component T_2 blocks: backbone total, side-chain total, aromatic total, nearest peptide C=O, and nearest peptide C–N. `mc_scalars.npy` stores an $N \times 6$ table with the four scalar sums $(f_{\mathrm{CO}}, f_{\mathrm{CN}}, f_{\mathrm{sidechain\,CO}}, f_{\mathrm{arom}})$ and the nearest peptide C=O and peptide C–N midpoint distances. If no accepted bond of one of those nearest-bond types is found, the stored distance remains 99.0 Å, the code's missing-value marker. Tensor and scalar values remain in Å^{-3}. Calibration against a DFT shielding reference is the later step that can attach physical scale and compare these descriptors with shielding changes.